

First-principles and thermodynamic nucleation modeling of hydrogen-driven vacancy evolution in silicon

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ABSTRACT

The performance of silicon-based devices is determined by crystal quality, which can be significantly affected by voids, the key intrinsic defects that influence electrical properties and dopant distribution. Therefore, controlling void concentration and size is crucial for optimizing silicon crystal growth and processing. Hydrogen incorporation can serve as a strategy to alter void evolution in Czochralski silicon by modifying the thermodynamic competition between surface energy and system entropy. Using first-principles calculations, we extract an intrinsic dangling-bond energy of 0.873 eV/bond and quantify the stepwise hydrogen binding energies on Si {111} void facets. By bridging these accurate first-principles parameters with a continuous surface energy formulation, we reveal that hydrogen adsorption gradually reduces the surface energy density, which leads to the inhibition of void growth. Thermodynamic analysis further demonstrates that direct hydrogen passivation effectively suppresses void growth by weakening the enthalpy driving force for vacancy aggregation and increasing the critical nucleation size x_c for stable void existence. Consequently, a unified thermodynamic framework describing the interaction between hydrogen and void surfaces is established, providing theoretical guidance for defect control in silicon crystal growth.

1. Introduction

Silicon crystal materials are the cornerstone of modern information society and the semiconductor industry. The performance of silicon-based devices critically depends on crystal quality, which is primarily governed by defects, with voids (bulk defects containing a large number of vacancies) being one of the most common type. These defects, especially in their larger forms, are associated with changes in electrical properties [1–3] and dopant distribution [4,5]. Therefore, controlling the concentration and size of voids has become an important topic in process optimization. Typically manifesting as an octahedral shape in the range of tens to hundreds of nanometers, voids form when vacancies agglomerate and grow under specific thermal histories [6–11]. In the Czochralski (Cz) silicon crystal growth, for example, high pulling rates yield vacancy supersaturation, forming retained octahedral agglomerates upon cooling [8,12].

In silicon crystal growth and processing, introducing hydrogen is proposed as a potential way to regulate the characteristics of vacancy-derived defects, as it can be trapped by these defects and thus modulate their subsequent growth behavior [13,14]. Studies have shown that high-temperature hydrogen annealing can effectively reduce the density of detectable voids in large-diameter Cz

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silicon [15]. This effect has been attributed to the regulatory role of hydrogen: Nakamura et al. [16], using simulation models in combination with the Gibbs–Thomson equilibrium equation and diffusion equations, demonstrated that hydrogen annealing establishes a zero oxygen concentration boundary condition at the wafer surface, which disrupts the structural stability of voids by inducing oxygen out-diffusion from the inner oxide layer of voids; Capaz et al. [17] revealed through ab initio molecular dynamics that hydrogen can significantly reduce the diffusion activation energy of oxygen in silicon, providing an efficient migration pathway for the released oxygen. These studies reveal that hydrogen can induce oxygen out-diffusion from voids and accelerate oxygen migration in silicon, thereby promoting the shrinkage or annihilation of voids.

Existing research has mainly focused on the mechanism by which hydrogen promotes oxygen diffusion and modifies the oxide layer on the inner walls of voids to eliminate defects [16,17]. However, the direct interaction between hydrogen and silicon surfaces within voids, not limited to those previously oxide-covered, remains relatively underexplored, as does its thermodynamic impact on void evolution. To bridge this gap and provide a complementary perspective, the present study establishes a theoretical framework to systematically explain the mechanism by which hydrogen inhibits void growth from a thermodynamic standpoint. By coupling first-principles calculations with classical nucleation theory, we derive an analytical formulation that explicitly links microscopic hydrogen coverage to macroscopic surface energy and critical nucleation size. This relationship enables us to translate atomic-scale adsorption energetics into a simple, practical thermodynamic criterion for void stability. Specifically, we quantify the energy gain from hydrogen adsorption and the trade-off between the energy benefit of surface area reduction and the penalty of configurational entropy loss. Our findings clarify why hydrogen intrinsically inhibits the further growth of octahedral voids and provide a concise thermodynamic method for predicting the variation trend of hydrogen-modulated nucleation thresholds in silicon defect engineering.

2. Methods

2.1. Ab initio computational details

All density functional theory (DFT) calculations were performed using the Vienna Ab initio Simulation Package (VASP) [18, 19]. The exchange–correlation effects were described utilizing the PBEsol generalized gradient approximation [20], while the ion–electron interactions were modeled via the projector augmented wave (PAW) method [21].

To ensure high accuracy in the defect calculations, a plane-wave cutoff energy of 450 eV was employed. The Brillouin zone was sampled using a $2 \times 2 \times 2$ Γ -centered k-point grid for the 216-atom supercell. For structures containing more than 6 vacancies, a 512-atom supercell with Γ -point only sampling was adopted. Structural relaxations were performed until the Hellmann–Feynman forces on each atom were less than 0.01 eV/Å. The plane-wave cutoff energy, k-point sampling, and supercell size were rigorously tested for convergence, as presented in Figure S1.

For transition state searches, considering the significant improvement of meta-GGA functionals in describing the energy surface, calculations were performed using the SCAN [22] functional. The migration barriers for the isolated vacancy and isolated hydrogen were directly obtained using the climbing image nudged elastic band (CI-NEB) method [23] with 3 intermediate images. Given the highly complex potential energy surface for coupled vacancy–hydrogen migration, a two-step strategy was implemented. First, the CI-NEB method was used to map the preliminary minimum energy pathway and provide an initial guess for the saddle point. Starting from the highest-energy image along the CI-NEB path, the dimer method [24] was then applied to precisely converge to the exact transition state and determine the accurate migration barrier.

Crystal orbital Hamilton population (COHP) analysis was conducted using the LOBSTER code [25] to analyze the stability of chemical bonding in the hydrogenated vacancies. The integrated COHP (ICOHP) up to the Fermi level was calculated to quantify how hydrogen adsorption modulates the relevant interactions at the adsorption site.

2.2. Binding and formation energies

Vacancy clusters (V_n) of varying sizes, ranging from the monovacancy ($n = 1$) to the ten-vacancy cluster ($n = 10$), were constructed by removing n silicon atoms from the lattice. The formation energy E_f can be expressed as [26]:

$$E_f = E_{\text{defect}} - E_{\text{bulk}} - \sum_i n_i \mu_i, \quad (1)$$

in which E_{defect} and E_{bulk} denote the total energies of the defect structure and the host supercell, respectively. Here, μ_i signifies the chemical potential of the i th species, while n_i represents the change in the number of atoms relative to the host supercell ($n_i < 0$ for removal and $n_i > 0$ for addition).

For hydrogenated vacancy clusters ($V_n H_m$), hydrogen atoms were introduced to saturate the dangling bonds (DBs). The geometric structures were fully relaxed to allow for surface reconstruction and the steric accommodation of adsorbed hydrogen atoms. For systems with multiple possible configurations, we performed configuration searches and structure relaxation calculations for all candidate structures individually, ultimately selecting the most stable configuration with the lowest energy as the energy reference for subsequent analysis.

To evaluate the thermodynamic stability of vacancy clusters under hydrogenation, we calculate the stepwise hydrogen binding energy, e_m . For a vacancy cluster $V_n H_{m-1}$ upon the adsorption of the m th hydrogen atom, this energy is defined as [27]:

$$e_m = E_f(V_n H_m) - E_f(V_n H_{m-1}) - E_f(\text{Si}_{\text{bulk}}\text{H}), \quad (2)$$

where $E_f(V_n H_m)$ is the formation energy of the defect supercell, and $E_f(\text{Si}_{\text{bulk}}\text{H})$ is the formation energy of a perfect silicon supercell containing one interstitial hydrogen atom.

2.3. Surface energy formulation

To quantify the energy changes associated with void aggregation and fragmentation in silicon crystals, we derived an analytical expression by mapping the atomistic concept of dangling bonds onto a continuous surface energy density.

The formation of a void involves the removal of silicon atoms from the perfect lattice, disrupting the continuous hybridized covalent network. This process leaves the silicon atoms at the void surface in an under-coordinated state, which gives rise to DBs. Thermodynamically, the creation of these DBs incurs a significant energy penalty, raising the total energy of the system relative to the perfect crystal. This excess energy constitutes the formation energy of the void.

Since these high-energy DBs are strictly localized at the void surface, under the condition where the total void volume is conserved (i.e., neglecting generation and annihilation), the total energy penalty scales linearly with the surface area. Therefore, we define the surface energy density γ as the excess energy per unit area.

Microscopically, γ is directly linked to the dangling bonds via the relation [28]:

$$\gamma \approx \sigma_{\text{DB}} \cdot E_{\text{DB}}, \quad (3)$$

where σ_{DB} is the areal density of dangling bonds (determined by the crystallographic orientation of the facet, e.g., Si{111}), and E_{DB} is the intrinsic energy cost of a single dangling bond. This framework allows us to translate the discrete, atomistic concept of “breaking bonds” into a continuous thermodynamic variable, providing a unified basis for analyzing void stability under different sizes and hydrogenation conditions.

The surfaces of the octahedral void are faceted by the {111} crystal plane of the silicon crystal. The dangling bond density σ_{DB} on Si{111} is:

$$\sigma_{\text{DB}\{111\}} = \frac{4}{\sqrt{3}a_0^2}, \quad (4)$$

where a_0 denotes the lattice constant of the standard cubic unit cell of silicon. Therefore, the surface energy density is:

$$\gamma_{\text{clean}} = \gamma_{\{111\}} = \sigma_{\text{DB}\{111\}} \cdot E_{\text{DB}}. \quad (5)$$

For a regular octahedron, l represents the edge length and the surface area is $A(l) = 2\sqrt{3}l^2$. The total surface energy term is:

$$E_{\text{surf}}(l) = A(l) \cdot \gamma_{\text{clean}} = 2\sqrt{3}l^2 \cdot \gamma_{\text{clean}}. \quad (6)$$

The energy change of the system upon adsorbing m hydrogen atoms is the cumulative sum of the stepwise hydrogen binding energies:

$$\Delta E_{\text{ads}}(m) = \sum_{i=1}^m e_i. \quad (7)$$

Therefore, the surface energy in the hydrogenated void with exact m hydrogen atoms is:

$$E_{\text{H-surf}} = E_{\text{surf}} + \Delta E_{\text{ads}} = A \cdot \gamma_{\text{clean}} + \sum_{i=1}^m e_i. \quad (8)$$

The effective surface energy density of the hydrogenated void, $\gamma_{\text{H}}(m)$ is:

$$\gamma_{\text{H}}(m) = \frac{E_{\text{H-surf}}}{A} = \gamma_{\text{clean}} + \frac{1}{A} \sum_{i=1}^m e_i. \quad (9)$$

The maximum number of available adsorption sites, N_{max} is determined by the intrinsic dangling bond density σ_{DB} of the Si{111} facets:

$$N_{\text{max}} = A \cdot \sigma_{\text{DB}}. \quad (10)$$

To quantify the effect of hydrogen passivation on void stability, we introduce the concept of normalized surface coverage, θ :

$$\theta = \frac{m}{N_{\text{max}}}. \quad (11)$$

Since hydrogen adsorption is limited by the number of dangling bonds (assuming mono-hydride termination), the coverage is physically constrained to the range: $0 \leq \theta \leq 1$. This relation $m = \theta N_{\text{max}}$ allows us to rewrite Eq. (9) explicitly as a function of coverage:

$$\gamma(\theta) = \gamma_{\text{clean}} + \frac{1}{A} \sum_{i=1}^{\theta N_{\text{max}}} e_i. \quad (12)$$

For large voids where N_{max} is large, the discrete summation can be approximated using an average adsorption energy per site, $\bar{e}(\theta)$:

$$\frac{1}{A} \sum_{i=1}^m e_i \approx \frac{m}{A} \cdot \bar{e}(\theta) = \frac{\theta N_{\text{max}}}{A} \cdot \bar{e}(\theta). \quad (13)$$

Using the relation $N_{\max}/A = \sigma_{\text{DB}}$, the expression simplifies to:

$$\begin{aligned}\gamma(\theta) &= \gamma_{\text{clean}} + \theta \cdot \sigma_{\text{DB}} \cdot \bar{e}(\theta) \\ &= [E_{\text{DB}} + \theta \cdot \bar{e}(\theta)] \cdot \sigma_{\text{DB}}.\end{aligned}\quad (14)$$

2.4. Thermodynamic equilibrium of voids

This section presents a thermodynamic analysis to quantify the stability of the fundamental growth steps of voids by calculating the Gibbs free energy change ($\Delta G = \Delta H - T\Delta S$) of voids during the process of successively adding vacancies: the capture of a monovacancy (V_1) by an existing void (V_x) to form V_{x+1} .

2.4.1. Enthalpy change: surface energy reduction

The driving force for vacancy aggregation is the reduction in total surface energy. The enthalpy change $d\Delta H$ is proportional to the change in total surface area:

$$d\Delta H/dx = \gamma \cdot [A(x+1) - A(x) - A(1)], \quad (15)$$

where $A(x)$ is the surface area of a void containing x vacancies. Assuming the void maintains a regular octahedral shape, the relationship between surface area and volume ($V_{\text{vac}} = xa_0^3/8$) yields the scaling law:

$$A(x) = C_{\text{geo}}x^{2/3}, \quad (16)$$

where the geometric prefactor is

$$C_{\text{geo}} = 2\sqrt{3} \left(\frac{3}{8\sqrt{2}} \right)^{2/3} \quad a_0^2 \approx 1.4298a_0^2. \quad (17)$$

Using the differential approximation when x is relatively large ($A(x+1) - A(x) \approx dA/dx$), the enthalpy change simplifies to:

$$d\Delta H/dx \approx -\gamma C_{\text{geo}} \left(1 - \frac{2}{3}x^{-1/3} \right). \quad (18)$$

2.4.2. Entropy change: configurational penalty

The aggregation process involves confining a dispersed vacancy from the bulk to a specific site on the void surface, resulting in a loss of configurational entropy. We assume the crystal is divided into unit boxes, where each box contains exactly one void. The volume of this box is determined by the void concentration:

$$V_{\text{box}} = \frac{1}{C_{\text{void}}}. \quad (19)$$

One box contains N vacancies distributed among M lattice sites. The number of microstates is:

$$\Omega_1 = \binom{M}{N} = \frac{M!}{N!(M-N)!}. \quad (20)$$

The configurational entropy is then given by $S_1 = k_B \ln \Omega_1$.

After adsorption, one vacancy is transferred from the lattice site to the void surface. Now there are $N-1$ vacancies distributed among M sites. The adsorbed vacancy has n_{surf} possible attachment sites on the surface of void, given by:

$$n_{\text{surf}} = A(x) \cdot \sigma_{\text{DB}} \approx (1.4298a_0^2x^{2/3}) \cdot \frac{4}{\sqrt{3}a_0^2} \approx 3.302x^{2/3}. \quad (21)$$

The number of microstates is the product of the bulk configuration and the surface possibilities:

$$\Omega_2 = \binom{M}{N-1} \times n_{\text{surf}} = \frac{M!}{(N-1)!(M-N+1)!} \cdot n_{\text{surf}}. \quad (22)$$

Analogously, the entropy is $S_2 = k_B \ln \Omega_2$. Since $M \gg N$ (lattice sites $\gg$ vacancies), we can approximate $M-N+1 \approx M$. Thus:

$$\frac{\Omega_2}{\Omega_1} \approx n_{\text{surf}} \cdot \frac{N}{M}. \quad (23)$$

Substituting the physical definitions $N = V_{\text{box}}C_V$ and $M = V_{\text{box}}\rho_{\text{Si}}$, where C_V is the vacancy concentration and ρ_{Si} is the atomic density of silicon, into Eq. (23) yields:

$$\frac{\Omega_2}{\Omega_1} \approx n_{\text{surf}} \frac{C_V}{\rho_{\text{Si}}}. \quad (24)$$

Therefore, the entropy change $d\Delta S$ is:

$$d\Delta S/dx = k_B \ln \left(\frac{\Omega_2}{\Omega_1} \right) \approx k_B \ln \left(n_{\text{surf}} \frac{C_V}{\rho_{\text{Si}}} \right). \quad (25)$$

This term is negative, representing an entropic barrier to growth.

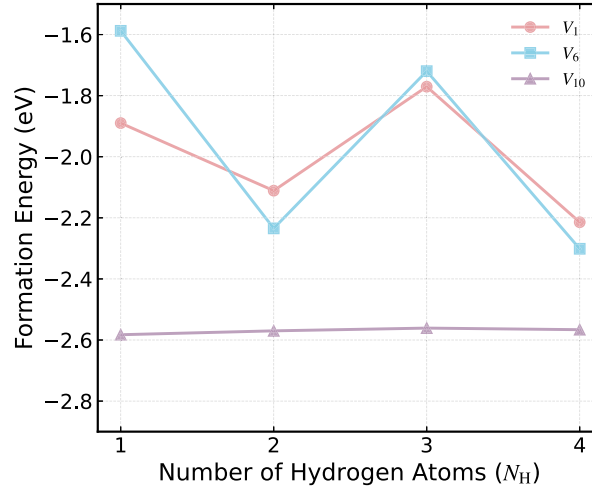


Fig. 1. Binding energies of vacancy clusters with varying numbers of H atoms. Each line represents a distinct vacancy cluster type, showing the energy evolution as the H atom number increases from 1 to 4.

2.4.3. Critical nucleation condition

In classical nucleation systems (e.g., solute clustering in alloys), the driving force (ΔG_{vol}) typically arises from the chemical potential difference between the supersaturated solid solution and the precipitating phase. In the case of vacancy aggregation in silicon, the process can be physically interpreted as an isolated vacancy entering a void, which reduces the number of dangling bonds and thus lowers the energy of the entire system. Therefore, in our specific derivation, the driving force is explicitly identified as the release of the energy of dangling bonds.

This physical mapping validates the enthalpy term used in Eq. (18). This highlights why the reduction of surface energy by hydrogen passivation is a factor in altering the critical size, as it directly lowers the energy gain from the reduction of dangling bonds during the vacancy aggregation process.

Substituting the enthalpy change from Section 2.4.1, Eq. (18), and the entropy change from Section 2.4.2, Eq. (25), into the thermodynamic equilibrium condition $\left. \frac{d\Delta G}{dx} \right|_{x=x_c} = 0$, we obtain the critical nucleation size (x_c):

$$-\gamma C_{geo} \left(1 - \frac{2}{3} x_c^{-1/3} \right) - k_B T \ln \left(n_{surf} \frac{C_V}{\rho_{Si}} \right) = 0. \quad (26)$$

Physically, x_c serves as the thermodynamic watershed determining the evolution of voids. It marks the dimension where the energy benefit from the reduction of total surface area counterbalances the penalty of entropy reduction. Voids smaller than x_c are unstable and prone to dissolution; those exceeding this size tend to grow spontaneously. Thus, x_c effectively acts as a critical threshold for void evolution, dictating the minimum void size required to stabilize the vacancy agglomerates.

3. Results and discussion

3.1. Dangling bonds and surface energy

We calculated the formation energies of a series of small vacancy clusters (V_n , $n = 1 \sim 10$) and adopted the average value derived from the 16 DBs of V_{10} (≈ 0.8730 eV) as the standard intrinsic dangling bond energy in our continuous formulation since V_{10} is the smallest of the V_n series to adopt an octahedral configuration.

In this work, the silicon lattice constant is taken as $a_0 = 5.4365$ Å. The dangling bond density σ_{DB} on Si{111} is:

$$\sigma_{DB} = \frac{4}{\sqrt{3}a_0^2} \approx 0.07814 \text{ Å}^{-2}. \quad (27)$$

The surface energy density is:

$$\gamma_{\{111\}} = \sigma_{DB} \cdot E_{bond} \approx 0.06821 \text{ eV/Å}^2. \quad (28)$$

The obtained value closely matches the experimental result of 0.077 eV/Å^2 [29]. The total surface energy term is:

$$E_{surf}(l) = (2\sqrt{3} \cdot \gamma_{\{111\}})l^2 \approx 0.2363l^2. \quad (29)$$

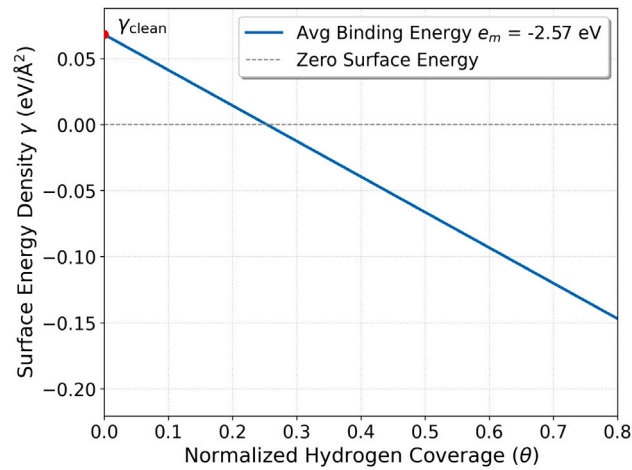


Fig. 2. Surface energy density shows a decreasing trend with increasing hydrogen coverage θ .

3.2. The effect of hydrogen on voids

By introducing n hydrogen atoms into the vacancy clusters and calculating their stepwise binding energies using Eq. (2), we obtained the results shown in Fig. 1. A noticeable contrast emerges in hydrogen binding behavior: while V_1 and V_6 exhibit pronounced odd–even oscillations in adsorption energy, V_{10} shows nearly constant binding energies across successive hydrogenation steps.

This “low-high” energetic pattern in V_1 and V_6 is a characteristic signature of the Jahn–Teller effect: in the pristine state, dangling bonds lower their energy by forming weakly reconstructed pairs via lattice distortion. The first hydrogen atom must overcome an energetic barrier to break this reconstruction. This energy cost partially offsets the gain from Si–H bond formation, resulting in a reduced effective binding energy. The first adsorption event leaves the neighboring silicon atom with an unpaired dangling bond, which is unstable. The second hydrogen atom attaches to this site without the need to break any further bonds. Consequently, it releases the full energy of the Si–H σ -bond formation, which corresponds to significantly strong binding.

However, in V_{10} , hydrogen atoms preferentially occupy the dangling bond sites located at the face centers of the vacancy cluster with a binding energy of approximately -2.57 eV. In voids, hydrogen atoms also tend to occupy such sites. Therefore, the binding energy of hydrogen in V_{10} serves as a representative parameter to characterize the reduction of surface energy by hydrogen.

As the void size increases, the local environment approaches that of a planar surface. To verify this, we constructed a surface slab structure and calculated the adsorption energy of hydrogen to be -2.59 eV, which is in excellent agreement with the value obtained in V_{10} . This validates the use of the adsorption energy at the V_{10} face center as a reliable approximation for the macroscopic surface.

Therefore, substituting $\bar{e}(\theta) = -2.57$ eV into the surface energy density equation:

$$\begin{aligned} \gamma(\theta) &= [E_{\text{DB}} - \theta \cdot \bar{e}(\theta)] \cdot \sigma_{\text{DB}} \\ &= [0.873 - 2.57 \cdot \theta] \cdot 0.07814 \text{ eV}/\text{\AA}^2. \end{aligned} \quad (30)$$

As illustrated in Fig. 2, the surface energy density decreases monotonically with increasing hydrogen coverage. Thermodynamically, the reduction in surface energy weakens the driving force for voids to merge, as the energy benefit of reducing the total surface area becomes less significant. This change alters the evolution of large voids: instead of continuous growth, their further expansion is suppressed.

Macroscopic continuous simulations [30], particularly recently advanced multiscale thermal-field modeling [12,31], provide powerful frameworks for evaluating void evolution. The analytical surface energy model $\gamma(\theta)$ developed here can be naturally integrated into these macroscopic approaches to incorporate the interfacial effects of hydrogen. For example, the constant surface energy γ in the classical nucleation barrier can be replaced by $\gamma(\theta)$, enabling standard nucleation-growth models to incorporate local chemical environments. Our work provides a thermodynamic link that complements existing macroscopic models and supports further investigation of hydrogen-modulated defect engineering.

Beyond modulating the macroscopic surface energy, hydrogen adsorption also significantly alters the local bonding environment. To elucidate this, we focus on the under-coordinated silicon atom (labeled Si0) neighboring the monovacancy V_1 , as shown in Fig. 3(a).

In a bare vacancy, the under-coordinated silicon atom forms a weak reconstructed Si–Si bond across the vacancy. When a hydrogen atom is introduced, it bonds with the silicon atom, forming a highly stable covalent Si–H bond. The $-ICOHP$ value of this new bond is significantly higher than that of the original weak reconstructed Si–Si interaction. Concurrently, the addition of hydrogen leads to a slight increase in the $-ICOHP$ values of the three adjacent Si–Si back-bonds (Fig. 3(b)). This demonstrates that hydrogen passivation causes the local environment of this silicon atom to more closely resemble a perfect silicon lattice. Consequently, the vacancy structure becomes more stable.

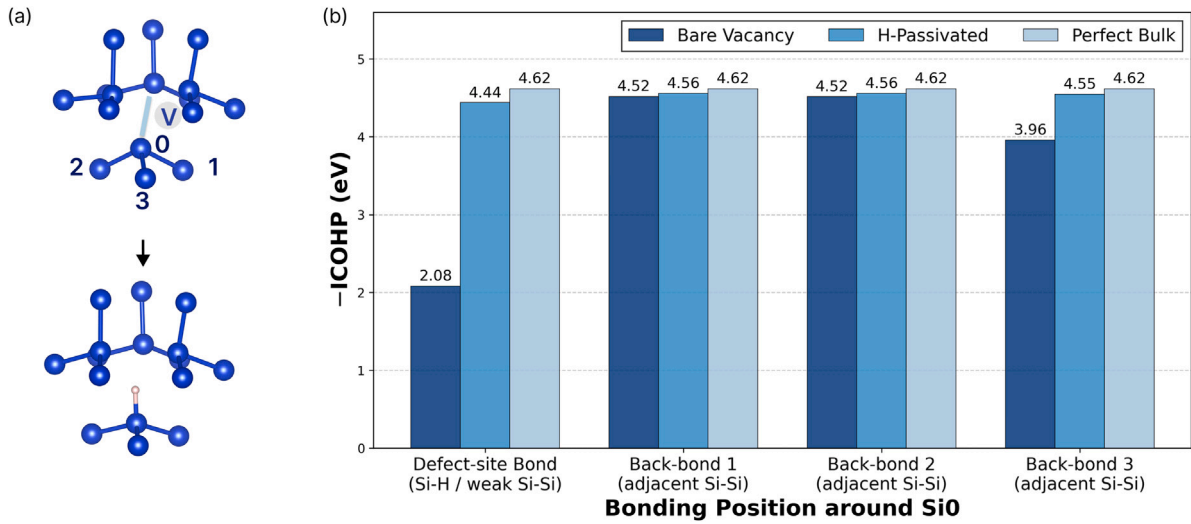


Fig. 3. (a) Geometry of the V_1 defect, illustrating the specific bonding orientations (connected to atom Si0) targeted for chemical bonding analysis. (b) Changes in calculated $-ICOHP$ values for various bonding environments. The “Defect-site Bond” group compares the weak reconstructed Si-Si interaction in the bare vacancy, the newly formed covalent Si-H bond in the passivated state, and the standard Si-Si bond in perfect bulk silicon. The remaining groups represent the adjacent Si-Si back-bonds, demonstrating structural relaxation and recovery toward the bulk state upon hydrogen adsorption.

This enhanced local stability substantially increases the kinetic difficulty for the vacancy to migrate through the lattice. To quantitatively verify this restricted mobility, we evaluated the migration barriers along the optimal kinetic pathways established in previous literature [32–34]. As shown in Fig. 4, both bare vacancies and isolated hydrogen atoms are highly mobile in the bulk silicon lattice, exhibiting low migration barriers of 0.41 eV and 0.27 eV, respectively. However, the formation of strong Si-H bonds effectively restricts the mobility of the under-coordinated silicon atoms, causing the migration barrier of the hydrogenated monovacancy to increase significantly to 1.29 eV. This transforms the highly mobile bare vacancy into a low-mobility complex. Since macroscopic void growth requires the rapid diffusion and agglomeration of vacancies, this hydrogen-induced kinetic constraint limits the supply of mobile vacancies, thereby inhibiting further void growth.

3.3. Critical nucleation size and multi-parameter analysis

Using the thermodynamic formulation established in Section 2.4, we further carried out a quantitative calculation of the critical nucleation size based on Eq. (26). Since the critical nucleation size depends on surface energy, which is reduced by hydrogen adsorption, the effect of hydrogen is naturally accounted for in our equations. The evolution of the critical size with temperature in this expression exhibits behavior consistent with thermodynamic expectations, thereby supporting the validity of the derived expression.

Substituting the standard values for silicon ($a_0 = 5.4365 \text{ \AA}$, $\gamma \approx 0.06821 \text{ eV/\AA}^2$) into Eq. (17), $\gamma C_{geo} \approx 2.882 \text{ eV}$, Eq. (26) becomes:

$$2.882 \left(1 - \frac{2}{3x_c^{1/3}} \right) + k_B T \ln \left(\frac{3.302x_c^{2/3} C_V}{\rho_{Si}} \right) = 0. \quad (31)$$

To systematically investigate the parameter dependence of the critical nucleation size x_c , we evaluated its numerical relationship with temperature T and hydrogen coverage θ , as shown in Fig. 5. The results presented in the figure correspond to the high-temperature regime of 1200–1600 K, combined with a moderate range of hydrogen coverages, where thermodynamic contributions are relatively significant [30,31]. By restricting our evaluation to this combined T and θ window, we preserve the validity of the equilibrium assumption: lower T would be increasingly governed by diffusion kinetics, while excessively high θ would lead to enhanced hydrogen-related atomic interactions beyond the scope of the current model. According to the derived equivalent surface energy relation, $\gamma(\theta) = [E_{DB} - \theta \cdot |\bar{e}(\theta)|] \cdot \sigma_{DB}$, the effective surface energy decreases rapidly with increasing θ . As shown in Fig. 5(a), the critical size x_c increases with θ across the high-temperature range. Fig. 5(b) further illustrates the T dependence. Cooling reduces x_c and promotes nucleation, in agreement with classical nucleation theory, while hydrogen systematically results in a higher x_c than in the pristine system ($\theta = 0$). This indicates that hydrogen-doped systems require larger voids to reach thermodynamic stability, effectively delaying the onset of void formation during the cooling process.

Fig. 5(c) maps the calculated x_c as a function of both θ and T . The contours represent the parameter space where voids are thermodynamically stable. At higher θ or T , x_c tends to approach infinity, indicating the thermodynamic instability of macroscopic

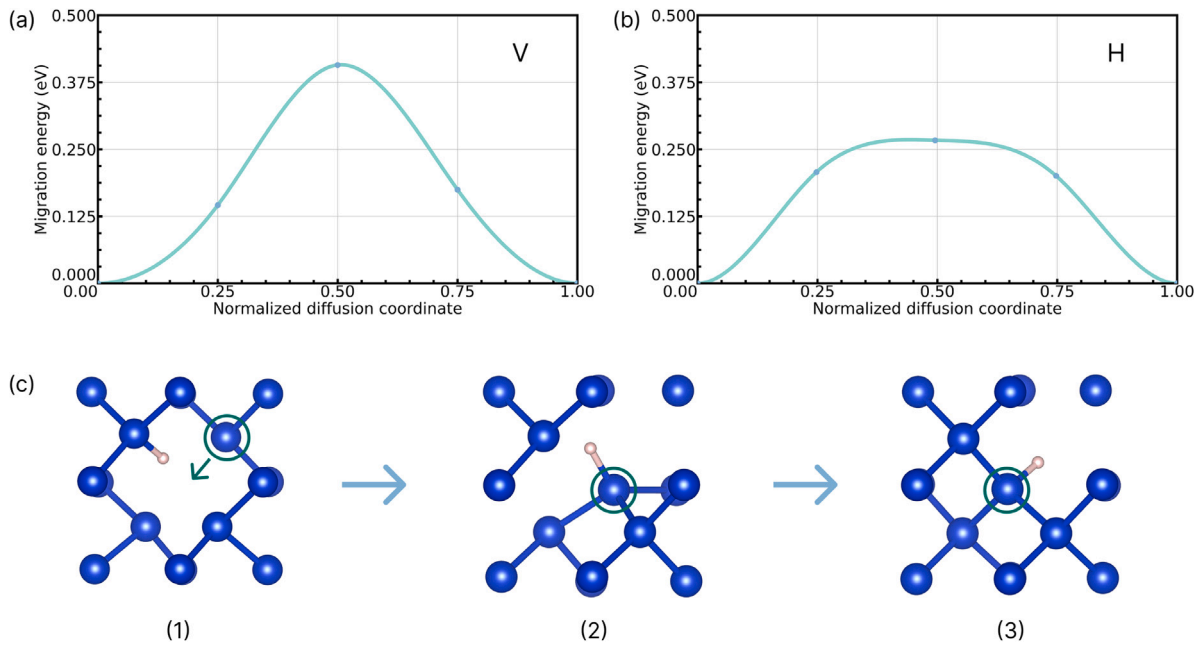


Fig. 4. Migration barriers and atomic pathways in silicon. (a) The migration energy of a bare monovacancy. (b) The migration energy of an isolated interstitial hydrogen atom moving between adjacent stable bond-center sites. (c) The coordinated diffusion pathway of a hydrogenated monovacancy (V_1H_1). The atomic configurations correspond to the initial state (1), the precise saddle point (2) identified via the dimer method, and the final state (3). Blue and pink spheres denote silicon and hydrogen atoms, respectively.

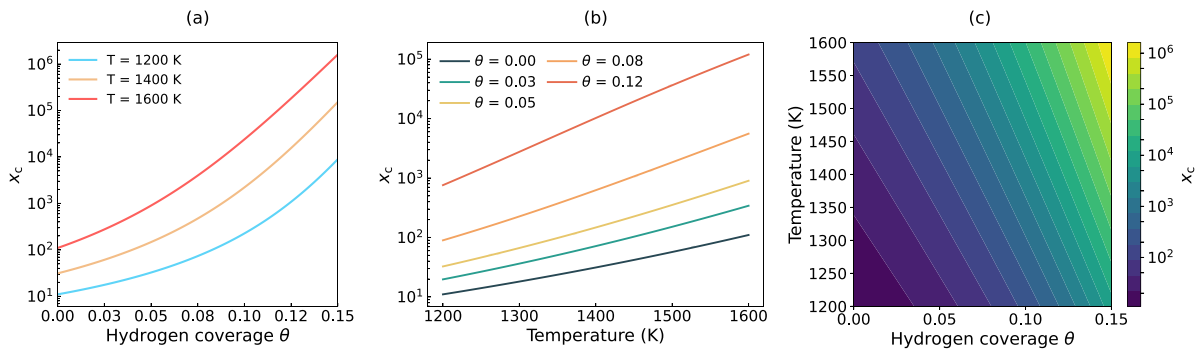


Fig. 5. Sensitivity analysis of the critical nucleation size x_c . (a) Effect of hydrogen coverage θ on x_c at various temperatures. (b) Temperature-controlled evolution of the critical size under distinct hydrogen coverages. (c) A 2D (θ, T) contour map showing the variation of the critical nucleation size x_c .

voids. Under these conditions, the entropic penalty for vacancy aggregation exceeds the enthalpy reduction provided by the surface energy. Consequently, continuous void aggregation becomes thermodynamically unfavorable, preventing void nucleation.

The overall trends and parameter sensitivities predicted by our theoretical framework are consistent with existing macroscopic experiments. For instance, Graf et al. [35] verified that high-temperature annealing of Cz silicon wafers in hydrogen ambients restricts the near-surface COP density and defect size. Ono et al. [36] demonstrated that pulling silicon single crystals in a hydrogen-bearing inert atmosphere results in grown-in COPs being either completely absent or drastically reduced in size. These findings directly echo our prediction that hydrogen narrows the thermodynamic stability window for regular voids.

4. Conclusion

In this study, we developed a theoretical framework to systematically elucidate the mechanism by which hydrogen modulates void evolution in silicon crystals from an indirect thermodynamic perspective. By bridging macroscopic thermodynamic equations with microscopic hydrogen adsorption parameters, we established a quantitative relationship between hydrogen coverage, surface energy, and critical nucleation size. Our findings reveal a dual synergistic mechanism for defect control: firstly, with hydrogen

coverage, the surface energy density drops, diminishing the tendency of voids to coalesce; secondly, during the initial aggregation stage, hydrogen adsorption significantly increases the critical nucleation size, thereby raising the thermodynamic barrier for void formation. These insights provide a unified analytical framework that directly connects atomic-scale chemical interactions to macroscopic void suppression, reducing the energy gain from vacancy aggregation and offering guidance for optimizing defect engineering in silicon crystal growth.

CRedit authorship contribution statement

Xinyi Zhong: Writing – original draft, Visualization, Methodology, Investigation, Formal analysis, Conceptualization. **Zirui He:** Methodology, Conceptualization. **Yuanchen Lyu:** Validation, Investigation. **Jiake Wang:** Investigation, Formal analysis. **Siqing Shen:** Writing – review & editing, Formal analysis. **Yongli Liang:** Writing – review & editing, Project administration, Conceptualization. **Hao Hu:** Writing – review & editing, Project administration, Conceptualization. **Shang-Peng Gao:** Writing – review & editing, Supervision. **Meng Chen:** Writing – review & editing, Supervision, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Appendix A. Supplementary data

Supplementary material related to this article can be found online at <https://doi.org/10.1016/j.micrna.2026.208732>.

Data availability

Data will be made available on request.

References

- [1] S. Huth, O. Breitenstein, A. Huber, U. Lambert, Localization of gate oxide integrity defects in silicon metal-oxide-semiconductor structures with lock-in IR thermography, *J. Appl. Phys.* 88 (2000) 4000–4003, <http://dx.doi.org/10.1063/1.1310185>.
- [2] J.-G. Park, G.-S. Lee, K.-D. Kwack, J.-M. Park, Crystal originated particle induced isolation failure in Czochralski silicon wafers, Japan. *J. Appl. Phys.* 39 (2000) 197, <http://dx.doi.org/10.1143/JJAP.39.197>.
- [3] J. Chen, D. Yang, Germanium doped Czochralski silicon, in: *Advances in Solid State Circuit Technologies*, 2010, pp. 367–392.
- [4] V. Raineri, P.G. Fallica, G. Percolla, A. Battaglia, M. Barbagallo, S.U. Campisano, Gettering of metals by voids in silicon, *J. Appl. Phys.* 78 (1995) 3727–3735, <http://dx.doi.org/10.1063/1.359953>.
- [5] S. Kilpeläinen, K. Kuitunen, F. Tuomisto, J. Slotte, E. Bruno, S. Mirabella, F. Priolo, Vacancy engineering by he induced nanovoids in crystalline si, *Semicond. Sci. Technol.* 24 (2008) 015005, <http://dx.doi.org/10.1088/0268-1242/24/1/015005>.
- [6] M. Isumi, M. Tomita, M. Yamawaki, The origin of defects in SiO₂ thermally grown on Czochralski silicon substrates, *J. Appl. Phys.* 78 (1995) 1940–1943, <http://dx.doi.org/10.1063/1.360232>.
- [7] J. Vanhellemont, Giant grown-in octahedral inclusions in as-grown silicon: voids versus silicon oxide precipitates, *Appl. Phys. Lett.* 69 (1996) 4008–4010, <http://dx.doi.org/10.1063/1.117853>.
- [8] G. Kissinger, G. Morgenstern, J. Vanhellemont, D. Gräf, U. Lambert, H. Richter, Internal oxidation of vacancy agglomerates in Czochralski silicon wafers during high-temperature anneals, *Appl. Phys. Lett.* 72 (1998) 223–225, <http://dx.doi.org/10.1063/1.120692>.
- [9] M. Isumi, Octahedral void defects in Czochralski silicon, *J. Cryst. Growth* 237–239 (2002) 1773–1778, [http://dx.doi.org/10.1016/S0022-0248\(01\)02337-5](http://dx.doi.org/10.1016/S0022-0248(01)02337-5).
- [10] M. Isumi, Grown-in defects in Czochralski silicon and their impact on gate oxide integrity, *Mater. Sci. Eng. B* 73 (2000) 184–190, [http://dx.doi.org/10.1016/S0921-5107\(99\)00458-4](http://dx.doi.org/10.1016/S0921-5107(99)00458-4).
- [11] M. Prasad, D. Sinno, Internally consistent approach for modeling solid-state aggregation. i. atomistic calculations of vacancy clustering in silicon, *Phys. Rev. B* 68 (2003) 045206, <http://dx.doi.org/10.1103/PhysRevB.68.045206>.
- [12] A.R.A. Dezfoli, Thermal history-driven void formation in silicon crystals: A discrete model approach, *Case Stud. Therm. Eng.* 73 (2025) 106468, <http://dx.doi.org/10.1016/j.csite.2025.106468>.
- [13] A.G. Ulyashin, R. Job, W.R. Fahrner, D. Grambole, F. Herrmann, Hydrogen redistribution and void formation in hydrogen plasma treated Czochralski silicon, *Solid State Phenom.* 82–84 (2001) 315–322, <http://dx.doi.org/10.4028/www.scientific.net/SSP.82-84.315>.
- [14] R. Job, A. Ulyashin, W. Fahrner, et al., Oxygen and hydrogen accumulation at buried implantation-damage layers in hydrogen- and helium-implanted Czochralski silicon, *Appl. Phys. A* 72 (2001) 325–332, <http://dx.doi.org/10.1007/s003390000622>.
- [15] D. Gräf, U. Lambert, M. Brohl, A. Ehlert, R. Wahlich, P. Wagner, Improvement of Czochralski silicon wafers by high-temperature annealing, *J. Electrochem. Soc.* 142 (1995) 3189, <http://dx.doi.org/10.1149/1.2048711>.
- [16] K. Nakamura, T. Saishoji, J. Tomioka, The dissolution behavior of the void defects by hydrogen annealing in Czochralski grown silicon crystals, *Appl. Phys. Lett.* 70 (1997) 3525–3527, <http://dx.doi.org/10.1063/1.119221>.
- [17] R.B. Capaz, L.V.C. Assali, L.C. Kimerling, K. Cho, J.D. Joannopoulos, Mechanism for hydrogen-enhanced oxygen diffusion in silicon, *Phys. Rev. B* 59 (1999) 4898–4900, <http://dx.doi.org/10.1103/PhysRevB.59.4898>.
- [18] G. Kresse, J. Furthmüller, Efficiency of ab-initio total energy calculations for metals and semiconductors using a plane-wave basis set, *Comput. Mater. Sci.* 6 (1996) 15–50, [http://dx.doi.org/10.1016/0927-0256\(96\)00008-0](http://dx.doi.org/10.1016/0927-0256(96)00008-0).
- [19] G. Kresse, J. Furthmüller, Efficient iterative schemes for ab initio total-energy calculations using a plane-wave basis set, *Phys. Rev. B* 54 (1996) 11169–11186, <http://dx.doi.org/10.1103/PhysRevB.54.11169>.
- [20] J.P. Perdew, A. Ruzsinszky, G.I. Csonka, O.A. Vydrov, G.E. Scuseria, L.A. Constantin, X. Zhou, K. Burke, Restoring the density-gradient expansion for exchange in solids and surfaces, *Phys. Rev. Lett.* 100 (2008) 136406, <http://dx.doi.org/10.1103/PhysRevLett.100.136406>.
- [21] P.E. Blöchl, Projector augmented-wave method, *Phys. Rev. B* 50 (1994) 17953–17979, <http://dx.doi.org/10.1103/PhysRevB.50.17953>.

- [22] J. Sun, A. Ruzsinszky, J.P. Perdew, Strongly constrained and appropriately normed semilocal density functional, *Phys. Rev. Lett.* 115 (2015) 036402, <http://dx.doi.org/10.1103/PhysRevLett.115.036402>.
- [23] G. Henkelman, B.P. Uberuaga, H. Jónsson, A climbing image nudged elastic band method for finding saddle points and minimum energy paths, *J. Chem. Phys.* 113 (2000) 9901–9904, <http://dx.doi.org/10.1063/1.1329672>.
- [24] G. Henkelman, H. Jónsson, A dimer method for finding saddle points on high dimensional potential surfaces using only first derivatives, *J. Chem. Phys.* 111 (1999) 7010–7022, <http://dx.doi.org/10.1063/1.480297>.
- [25] R. Nelson, C. Ertural, J. George, V.L. Deringer, G. Hautier, R. Dronskowski, Lobster: Local orbital projections, atomic charges, and chemical-bonding analysis from projector-augmented-wave-based density-functional theory, *J. Comput. Chem.* (2020) 1–10.
- [26] A. Goyal, P. Gorai, H. Peng, S. Lany, V. Stevanović, A computational framework for automation of point defect calculations, *Comput. Mater. Sci.* 130 (2017) 1–9, <http://dx.doi.org/10.1016/j.commatsci.2016.12.040>.
- [27] T. Ozaki, C.-C. Lee, Absolute binding energies of core levels in solids from first principles, *Phys. Rev. Lett.* 118 (2017) 026401, <http://dx.doi.org/10.1103/PhysRevLett.118.026401>.
- [28] S.B. Zhang, S.-H. Wei, Surface energy and the common dangling bond rule for semiconductors, *Phys. Rev. Lett.* 92 (2004) 086102, <http://dx.doi.org/10.1103/PhysRevLett.92.086102>.
- [29] D.J. Eaglesham, A.E. White, L.C. Feldman, N. Moriya, D.C. Jacobson, Equilibrium shape of Si, *Phys. Rev. Lett.* 70 (1993) 1643–1646, <http://dx.doi.org/10.1103/PhysRevLett.70.1643>.
- [30] T. Sinno, R. Brown, Modeling microdefect formation in Czochralski silicon, *J. Electrochem. Soc.* 146 (1999) 2300–2312, <http://dx.doi.org/10.1149/1.1391931>.
- [31] S.-K. Yang, A.R.A. Dezfoli, Multiscale modeling of cop suppression in Czochralski grown silicon: Insights into cooling tube effects, *Mater. Sci. Semicond. Process.* 200 (2025) 109923, <http://dx.doi.org/10.1016/j.mssp.2025.109923>.
- [32] P. Deák, L.C. Snyder, J.W. Corbett, State and motion of hydrogen in crystalline silicon, *Phys. Rev. B* 37 (1988) 6887–6892, <http://dx.doi.org/10.1103/PhysRevB.37.6887>.
- [33] M.A. Roberson, S.K. Estreicher, Vacancy and vacancy-hydrogen complexes in silicon, *Phys. Rev. B* 49 (1994) 17040–17049, <http://dx.doi.org/10.1103/PhysRevB.49.17040>.
- [34] S.K. Estreicher, A. Docaj, M.B. Bebek, D.J. Backlund, M. Stavola, Hydrogen in c-rich si and the diffusion of vacancy–h complexes, *Phys. Status Solidi A* 209 (2012) 1872–1879, <http://dx.doi.org/10.1002/pssa.201200054>.
- [35] D. Gräf, U. Lambert, M. Brohl, A. Ehlert, R. Wahlich, P. Wagner, Comparison of high temperature annealed czochralski silicon wafers and epitaxial wafers, *Mater. Sci. Eng. B* 36 (1996) 50–54, [http://dx.doi.org/10.1016/0921-5107\(95\)01265-6](http://dx.doi.org/10.1016/0921-5107(95)01265-6).
- [36] T. Ono, W. Sugimura, M. Hourai, Silicon wafer, method for producing silicon wafer and method for growing silicon single crystal, 2009.